

Portfolio

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Simulating visual imagination in a memory model

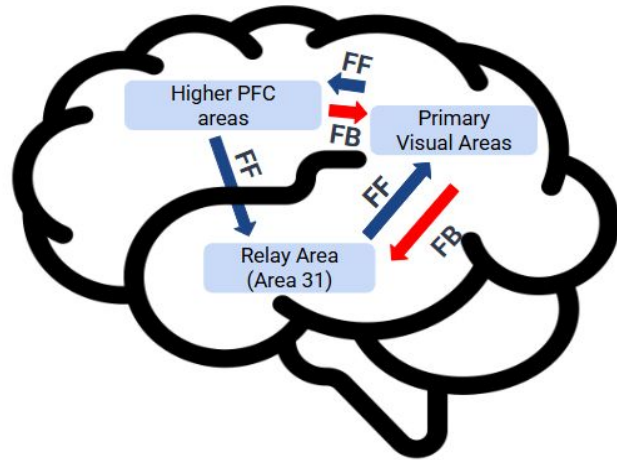


Fig. 1. The pathway of visual imagination

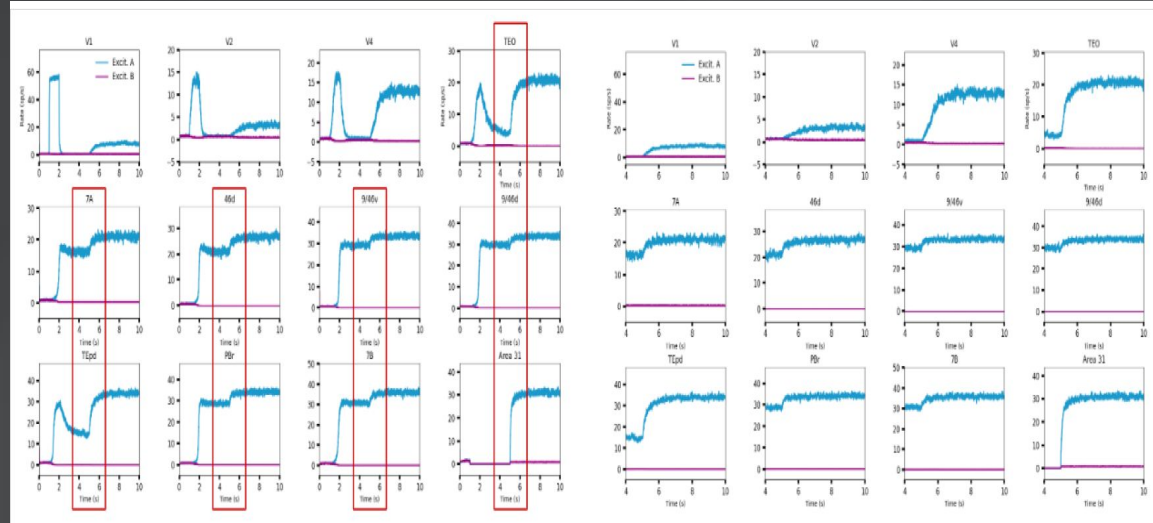


Fig. 2. Activity in the new brain area where visual imagination occurs

This project explores how brain generates images by using a simulation model of memory in monkeys. I added a new brain region to simulate communication between the prefrontal and visual areas. This helped to understand how different parts of the brain work together during visual imagination. The model uses real brain connectivity data to create a biologically realistic simulation.

Model: Wong-Wang working memory model
Programming: Python
Packages: Numpy, Matplotlib, Pandas

Visual Search Task Experiment

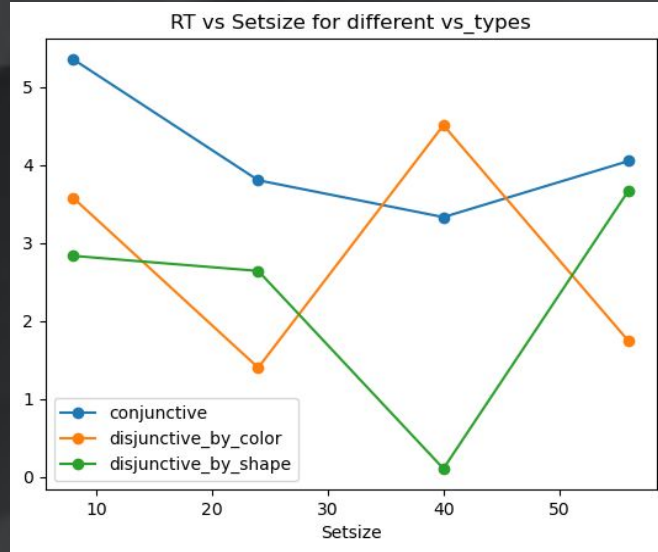


Fig. 1. Mean of reaction times for every condition

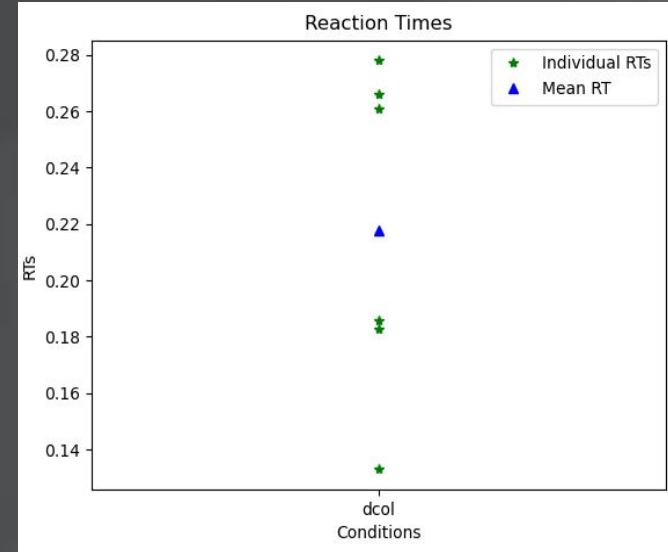


Fig. 2. Reaction time of a subject compared to the mean

This project implements a Visual Search Task (VST) based on Feature Integration Theory (FIT) to study how people identify targets in visual environments. Participants search for an odd symbol among distractors in three conditions: disjunctive by color, disjunctive by shape, and conjunctive search, along with a control task.

Programming Language: Python
Packages: NumPy, Pandas, Matplotlib

Shape Flanker Task Experiment

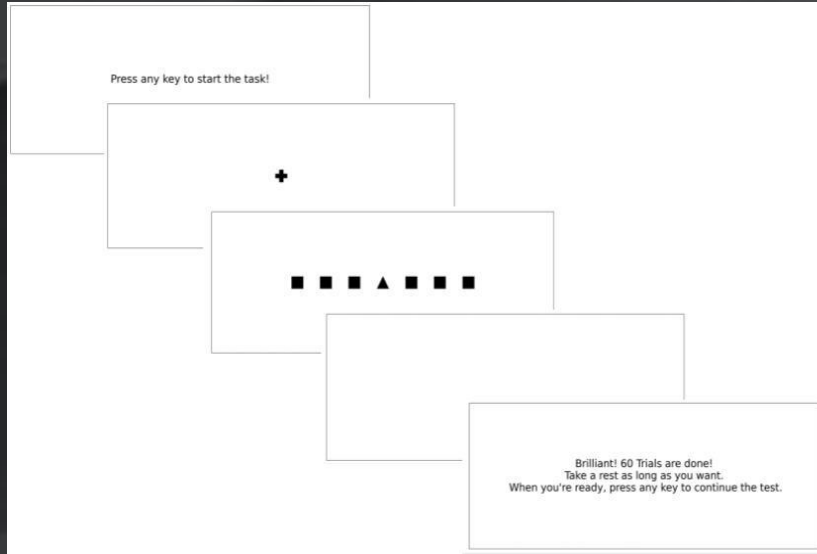


Fig. 1. Frames of the progression of the experiment

This project explores selective attention and inhibitory control using a modified Flanker Task. The study examines if the Flanker Effect (slower responses when surrounded by different objects) works the same with shapes as it does with arrows or letters. The participants respond to target shapes (triangles or squares) while ignoring surrounding distractors.

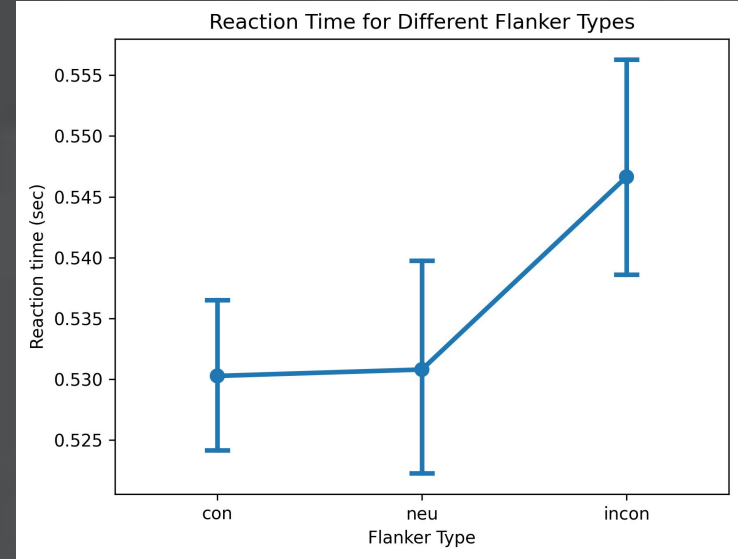


Fig. 2. ANOVA of mean reaction times per condition

Programming Language: Python
Packages: NumPy, Pandas, Matplotlib
Design: Object-Oriented Programming

EEG Study

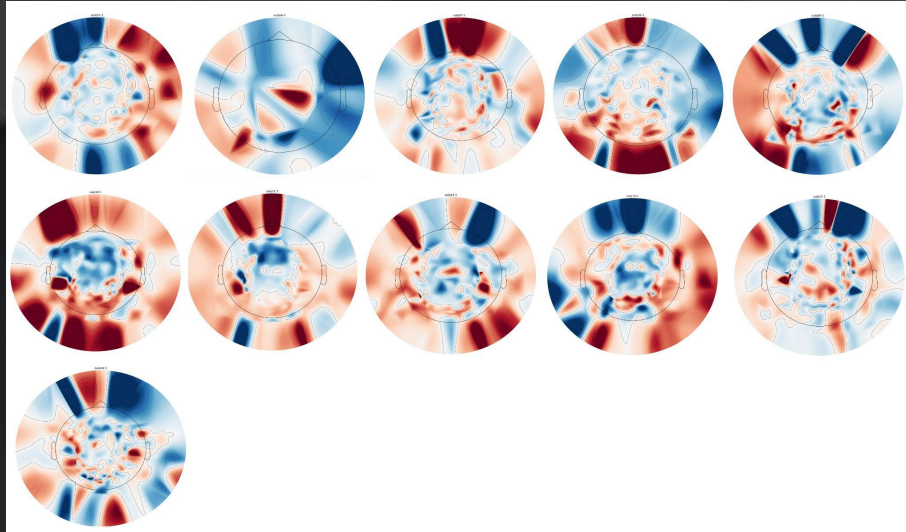


Fig. 1. Topological plots of brain activity

This project studies how fast people process visual information when making decisions. I used EEG to observe brain activity in response to a visual stimulus, focusing on its activity around 200ms. The information was estimated through Singular Value Decomposition (SVD) and analyzed it using linear and Bayesian regression models.

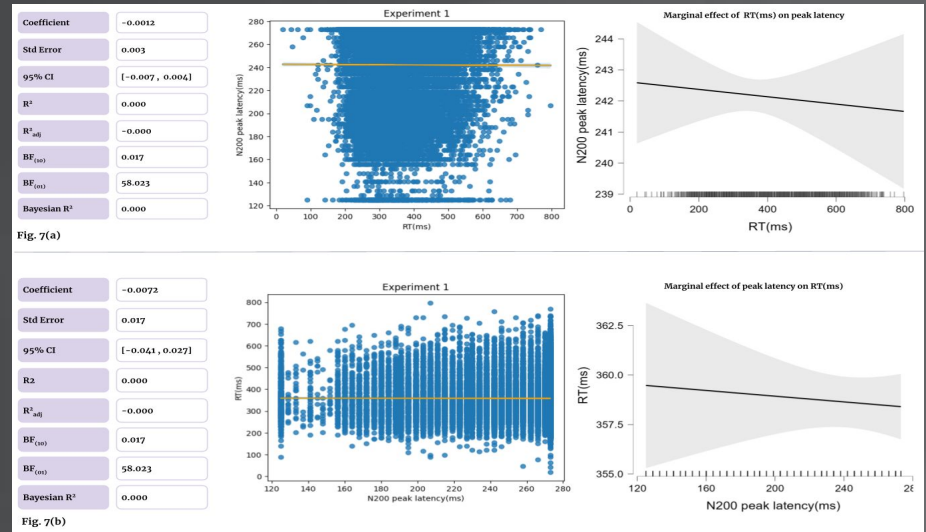


Fig. 2. Relationship between reaction time and brain activity

Programming Language: Python
Packages: MNE, NumPy, Pandas, Statsmodels, Matplotlib, Seaborn
Software: JASP
Signal Processing: ICA (Independent Component Analysis), SVD (Singular Value Decomposition)

Neuron Simulation Study

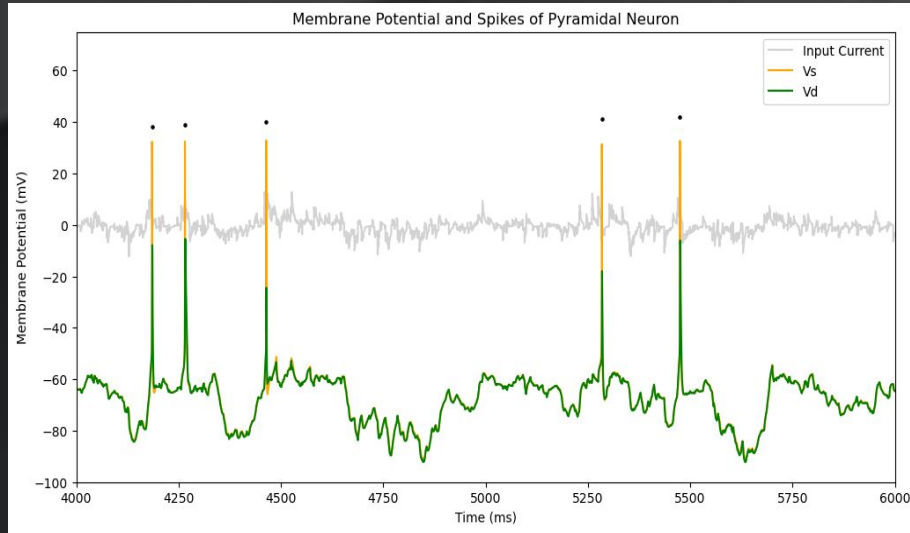


Fig. 1. Simulated Neuron Activity

This project uses computer simulations to understand how the communication speeds between neurons influence their collective behavior. I created a virtual model of two connected neurons and varied their speed of communication to observe its responses.

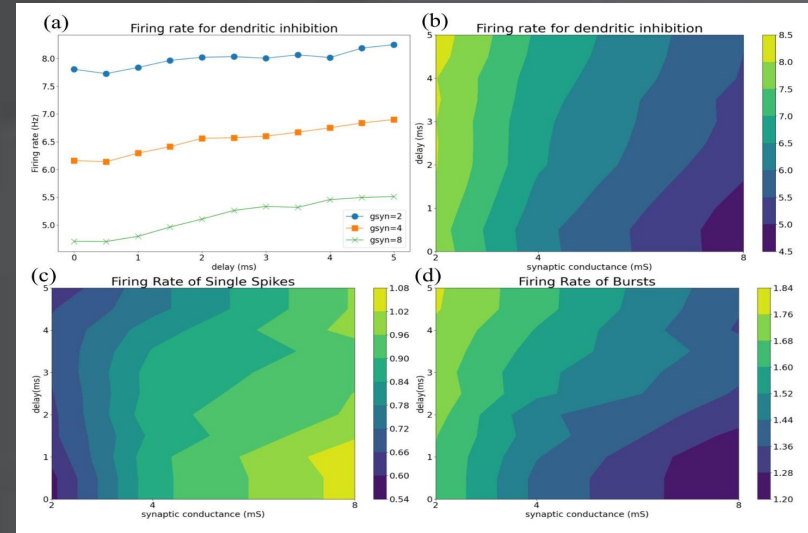


Fig. 2. Firing rate of neurons after adding a delay in their connection

Programming: Python

Data Analysis & Signal Processing: Power spectral analysis, inter-spike interval distributions, cross-correlations

Packages: NumPy, SciPy, Matplotlib, Brian Simulator

Neuron Model: Pinsky-Rinzel model, Wang-Buzsáki model